

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

2bi2____ / 2IPbi2__

REGISTRATION NUMBER

BIOLOGY

9648/03

Paper 3

31 August 2016

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

Sections A - C

Answer **all** questions in the spaces provided on the question paper.

Sections D - E

Answer **all** questions on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 14
2	/ 14
3	/ 12
4	/ 12
5	/ 20
TOTAL	/ 72

This document consists of **11** printed pages.

[Turn over

Section A

Answer the question in this section.

- 1 In a maternity ward at a local hospital, a mix-up involving three couples and three babies caused a lot of confusion. Based on phenotypic characteristics, the nurses were unable to correctly identify the parents of the babies. In order to solve the case, a scientist was called in to carry out a DNA test to identify the parents of the babies. The test was based on the principle that different individuals have a different number of repeating units at a particular locus in a chromosome.

Chromosome 13 was isolated from the DNA samples that were obtained from the three couples and three babies and used for further analysis. The sequence below shows a segment of chromosome 13, which was used in the analysis where (TTAGGAT) is the repeating unit and n is the number of repeats.

5' ...GCTAAGTATTGCTCAAGA... (TTAGGAT)n...GATAAATAACTGGCTAGTA...-3'
 3' ...CGATTCATAACGAGTTCT... (AATCCTA)n... CTATTTATTGACCGATCAT...-5'

Fig. 1.1 shows the results of the DNA test obtained from each individual.

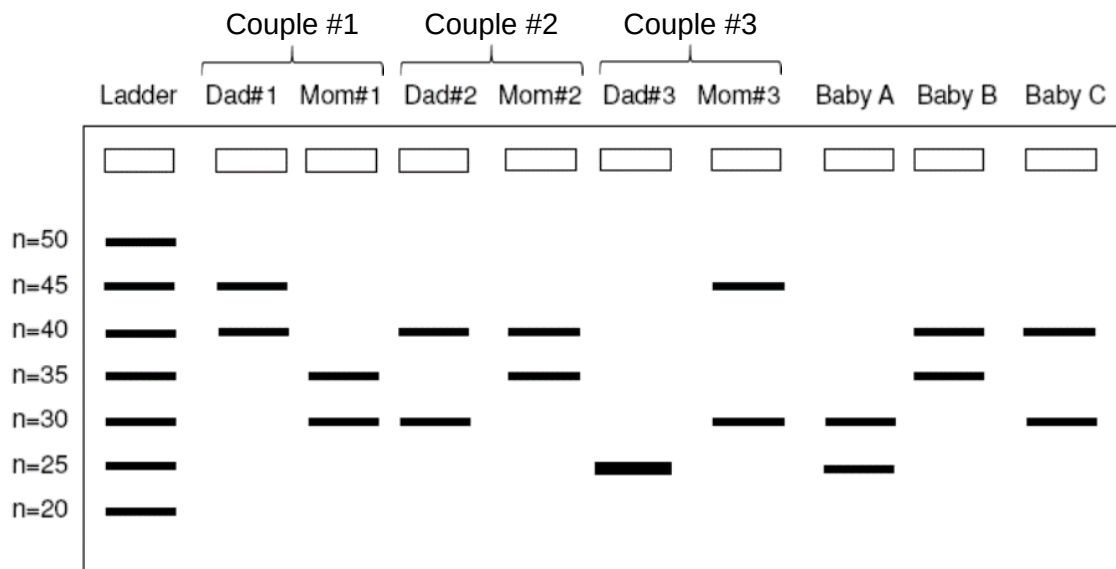


Fig.1.1

- (a) Describe how the DNA bands in the gel could be made visible.

Staining with ethidium bromide and viewing under UV light

OR

Staining with methylene blue

[1]

- (b) State the purpose of the DNA ladder.

Each band serves as a comparison / reference for the size / number of repeats present in the individuals tested.

[1]

(c) Explain how gel electrophoresis is used to separate fragments of DNA.

1. Because all DNA molecules are negatively charged, regardless of the length or source, the rate of DNA migration and separation through an agarose gel depends on the size / molecular length of the DNA molecule.
2. An agarose gel is submerged in a buffer solution containing ions that will conduct electricity.
3. DNA samples are loaded into small depressions in the gel called wells, which are close to the negative electrode / cathode.
4. A direct current is applied through electrodes at opposite ends of the gel.
5. The negatively charged DNA molecules move toward the positive electrode / anode, with shorter DNA molecules moving faster and further than the longer ones.

[5]

(d) Explain the banding pattern of Dad #3.

(Any 3)

1. Only one band corresponding to $n=25$
2. Band is twice as thick as the other bands
3. Homozygous for the locus being examined
4. 25 repeats on both copies of chromosome 13

[3]

(e) Identify the couple that Baby A belongs to.

Couple #3

[1]

(f) Explain why the results shown in Fig. 1.1 could not confirm which couple that Baby B belongs to.

(Any 2)

1. Each band in Baby B's DNA fingerprint would match the band in either the mum's or dad's DNA fingerprint.
2. The band corresponding to $n=40$ in Baby B's DNA fingerprint can be found in Dad#1, Dad#2 and Mom#2, whereas the band corresponding to $n=35$ in Baby B's DNA fingerprint can be found in Mom#1 and Mom#2.
3. Hence, Baby B could belong to either Couple #1 or Couple #2.

[2]

- (g) Suggest an alternative DNA test to identify the couple that each of the three mixed up babies belongs to.

Analysis of Restriction Fragment Length Polymorphism (RFLP) or other Short Tandem Repeats (STR) [1]

[Total: 14]

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Section B

Answer the question in this section

- 2 Haemophilia is an X-linked recessive disorder that impairs the body's ability to form blood clots. There are two main types, haemophilia A and haemophilia B, resulting from deficiencies in clotting factors VIII and IX respectively.

Both *ex vivo* and *in vivo* gene therapy approaches are undergoing clinical trials.

- (a) (i) Explain why gene therapy approach for treatment of haemophilia is possible.

.....

[1]

- (ii) Distinguish between *ex vivo* and *in vivo* gene therapy approaches.

.....

[2]

Haematopoietic stem cells (HSC) are ideal vehicles for *ex vivo* gene therapy application.

- (b) (i) Describe the normal function of HSC.

.....

[1]

- (ii) Explain why HSC are ideal vehicles for *ex vivo* gene therapy application.

.....

[3]

The liver is an ideal target for *in vivo* gene therapy application. As a major organ and central metabolic hub, it receives an abundant blood supply through sinusoids with highly permeable walls, which facilitates easy access of blood-borne particles to the hepatocytes. Hepatocytes are long-lived and robust protein factories that can efficiently release their products into the blood circulation. Both viral and non-viral vectors have been and continued to be investigated.

- (c)** Use of non-viral vectors for gene therapy has met with low success rate.

Describe one method of non-viral vector delivery and explain why the method has achieved low success.

[3]

- (d)** Many ongoing clinical trials are focused on the application of adeno-associated viral (AAV) vectors, which are found to remain as episomes within the nucleus.

Explain an advantage and a disadvantage of the use of such adeno-associated viral vectors compared to retroviral vectors.

[2]

- (e) Somatic cell nuclear transfer (SCNT), also known as therapeutic cloning, involves the replacement of an egg nucleus with a somatic cell nucleus. As the oocyte develops into a blastocyst, cells from the inner cell mass can be isolated and purified to serve as a source of pluripotent stem cells.

Explain why such an approach cannot be used to treat patients with haemophilia.

[2]

[Total: 14]

- 2 Haemophilia is X-linked recessive disorder that impairs the body's ability to form blood clots. There are two main types, haemophilia A and haemophilia B, resulting from deficiencies in clotting factor VIII and IX respectively.

Both *ex vivo* and *in vivo* gene therapy approaches are undergoing clinical trials.

- (a) (i) Explain why gene therapy approach for treatment of haemophilia is possible. [1]
 1. Haemophilia is caused by single defective gene.
- (ii) Distinguish between *ex vivo* and *in vivo* gene therapy approaches. [2]
 1. In *ex vivo* gene therapy, the therapeutic gene is introduced into target cells extracted from patient, and the altered cells are injected back into the patient.
 2. In *in vivo* gene therapy, the therapeutic gene is introduced directly into target cells/tissues in the patient's body.

Haematopoietic stem cells (HSC) are ideal vehicles for *ex vivo* gene therapy application.

- (b) (i) Describe the normal function of HSC. [1]
 1. HSC normally differentiates to give rise to many types of blood cells, such as red blood cells, white blood cells and platelets.
- (ii) Explain why HSC are ideal vehicles for *ex vivo* gene therapy application. [3]
 1. HSC can undergo indefinite self-renewal, hence they can multiply indefinitely, thus sustaining the gene therapy treatment.
 2. HSC are unspecialised

3. HSC have potential to differentiate, they can be stimulated to differentiate to produce the therapeutic gene product
4. HSCs multiply themselves $>10^6$ fold during haematopoietic reconstitution
5. HSCs can be manipulated to secrete biotherapeutic molecules such as FVIII directly into the bloodstream
6. HSCs can induce a state of immune tolerance or nonresponsiveness to the therapeutic transgene product

Accept:

A1. Given that HSCs are derived from patient, there is lower risk of immune rejection.

Reference: Spencer et.al. (2016). State of the art; gene therapy of haemophilia HSCs are ideal cellular vehicles for gene therapy applications since they can (i) self-renew, (ii) multiply themselves $>10^6$ fold during haematopoietic reconstitution, and (iii) secrete biotherapeutic molecules such as FVIII directly into the bloodstream. Another critical property of HSC-directed gene therapy is the ability to induce a state of immune tolerance or nonresponsiveness to the therapeutic transgene product, which in the case of haemophilia A is a single protein, FVIII, that is known to possess a higher degree of immunogenic potential.

Reject:

(R!) HSCs can be isolated/ culture easily

(R!) lower risk of tumor formation compared to using pluripotent cells

The liver is an ideal target for *in vivo* gene therapy application. As a major organ and central metabolic hub, it receives an abundant blood supply through sinusoids with highly permeable walls, which facilitates easy access of blood-borne particles to the hepatocytes. Hepatocytes are long-lived and robust protein factories that can efficiently release their products into the blood circulation. Both viral and non-viral vectors have been and continue to be investigated.

(c) Use of non-viral vectors for gene therapy has met with low success rate.

Describe one method of non-viral vector delivery and explain why the method has achieved low success.

[3]

Naked DNA method

- L1. Therapeutic DNA is injected directly into target tissues.
- L2. Target cells take up DNA randomly
- L3. Poor transfection efficiency
- L4. Not integrated into genome, hence transient expression

Liposomes

- N1. Therapeutic DNA is encapsulated in anionic liposome.
- N2. Mixing of liposomes and target cells result in fusion, thus introducing gene into target cell
- N3. Poor transfection efficiency
- N4. Not integrated into genome, hence transient expression

Molecular conjugates

- M1. Therapeutic DNA is coupled to a targeting molecule.

- M2.Targeting molecule binds to specific cell surface receptor and induces endocytosis and transfer of DNA into cells.
- M3.Molecular conjugate often remain trapped in endosome resulting in poor gene transfer
- M4.Not integrated into genome, hence transient expression.

- (d) Many ongoing clinical trials are focused on the application of adeno-associated viral (AAV) vectors, which are found to remain as episomes within the nucleus.

Explain an advantage and a disadvantages of use of such adeno-associated viral vectors compared to retroviral vectors. [2]

Advantages:

- A1.AAV vectors does not result in integration of therapeutic DNA into host genome unlike retroviral vectors hence less likely to cause insertional mutagenesis.
- A2.AAV vectors readily infects both dividing and non-dividing cells, unlike retroviral vectors which can only infect dividing cells

Disadvantages

- D1.AAV vectors does not result in integration of therapeutic DNA into host genome unlike retroviral vectors hence expression may be transient.
- D2.AAV vectors can only accept smaller insert size, up to 4.5kb, compared to retroviral vectors which can accepts to larger insert size of up to 8 kb.
- D3.AAV vectors require helper virus to infect target cells, while retroviral vectors do not.

Reject:

Advantage:

- 1. AAV vectors have higher transfection efficiency. (A!)
- 2. AAV vectors can target specific tissues and organs, which is not the case for retroviral vectors. (A!)
- 3. AAV vectors elicits lesser immune response (A!)
- 4. AAV vectors less likely to regain viral properties than retroviral vectors (A!)

Disadvantage

- 5. AAV vectors have lower transfection efficiency. (A!)
- 6. Less familiarity with AAV vectors, thus potential danger of AAV vectors is unknown. (R!)
- 7. DNA insert in AAV vectors is not inserted into cell genome, hence level of DNA expression is significantly lower. (A!)
- 8. Repeated therapy cycles may be needed hence more costly.

- (e) Somatic cell nuclear transfer (SCNT), also known as therapeutic cloning, involves the replacement of an egg nucleus with a somatic cell nucleus. As the oocyte develops into a blastocyst, cells from the inner cell mass can be isolated and purified to serve as a source of pluripotent stem cells.

Explain why such as approach cannot be used to treat patients with haemophilia. [2]

- 1. All somatic cells from the patient will contain the defective gene.
- 2. The resulting pluripotent cells isolated from inner cell mass from such a procedure will contain the same genetic defect, hence remain unable to produce the necessary clotting factors.
- 3. If a donor cell nucleus containing a functional gene is used for SCNT, the stem

cells may contain antigens that cause them to be rejected by the immune system.

[Total: 14]

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Section C

Answer the question in this section.

- 3 The success of culturing callus is low due to the difficult task of removing contaminants. It has proven more challenging for plants taken from the wild in tropical countries. In Sumbawa, Indonesia, plant tissue samples were sampled at three different times of the year. They were grown in medium containing no fungicide or antibiotic.

Table 3.1 shows the results.

Table 3.1

explant	time of year	number of explants	number of cultured explants with no fungal or bacterial contamination	percentage of cultured explants with no fungal or bacterial contamination
leaf disc	April	153	12	8
	August	322	16	5
	January	332	30	9
shoot tip	April	194	116	60
	August	191	122	64
	January	211	156	74

- (a) Describe and explain the results of using the two types of explant.

Using shoot tip had higher rate of success;

Where for example in April, using shoot tip explant produced 60% cultured

explants with no contamination compared to the 8% of uncontaminated explants that

were produced when leaf discs were used; (A! any appropriate comparison of values

in the months of august or January) OR

Explant taken during January had a higher success rate with 8% for leaf disc

and 74% for shoot tip;

Shoot tip contains meristematic cells which are actively dividing;

[3]

Use antibiotics/fungicide;

Use of dilute sodium hypochlorite/ bleach to sterilise wild plant samples;

[1]

- (b) Besides using the appropriate explant, suggest how the number of contaminated samples could be reduced when using wild plant samples.

- (c) Explain, with relevant examples, how genetic engineering has helped to increase the quantity of crops for farmers.

Any relevant technology;

Change;

Benefit to farmers;

E.g. Super Salmon à promoter for AFP plus beside the gene coding for growth;

Gene will be activated all year round due to cold environment and growth hormone year round;

Salmon can grow twice as fast with same amount of feed;

[3]

- (d) Describe two disadvantages of plant tissue culture.

Contamination of cultures poses the greatest problem to commercial tissue culture as it can cause very high losses in a short time;

Micropropagation is tedious and costly as it requires much labour (e.g transfer of plantlets from the laboratory to the soil), trained personnel with specialized skills, sophisticated facilities and organization, sterile laboratory conditions and special nutrient media. This may not be economical for crops with low financial returns like carrots;

Plants produced from calli may undergo genetic changes to produce genetic off-types. For example, bananas can produce a lot of genetic off-types in culture. Most of these changes are undesirable;

The limited genetic pool and genetic uniformity of plants cultured make them

vulnerable to new diseases or drastic changes in the environment;

[Any two points]

[2]

- (e) In several countries including China, Korea and the United States, human DNA has already been put into eggs from both rabbits and cows.

Discuss the ethical concerns of conducting such experiments.

Humans should not be tampering with nature by creating living organisms which may

be objectionable by certain religious groups;

Crossing the species barrier and violating the genetic integrity of the organism;

The research may not be justified if it is not to address the urgent needs of mankind

only the desires of mankind;

Exploitation of animals in the process/ Lack of concern for their welfare;

Slippery slope that may lead to human cloning; AVP

[3]

Section D

Answer the question in this section on the answer paper provided.

4 Planning Question

You are required to plan, but not carry out, an investigation into the effect of increasing concentration of glucose on rate of respiration of yeast.

Yeast synthesizes ATP through two major biochemical pathways: aerobic respiration and fermentation. During both aerobic respiration and fermentation, yeast cells break down glucose molecules within the cell to release energy, and some of this energy is captured and stored in the ATP's high-energy phosphate bonds. The breakdown of glucose also releases carbon atoms, which become available for biosynthetic reactions, enabling the yeast to grow and reproduce by budding. The rest of the carbon ends up in the by-products of these reactions.

You are to use methylene blue in your investigation. Methylene blue acts as an artificial electron acceptor during respiration, which changes from blue to colourless as a result of its reduction in the enzymatic reaction.

Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure that they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- yeast suspension,
- 10% glucose solution,
- distilled water,
- access to tap water,
- thermostatically controlled water bath,
- methylene blue,
- stopwatch,
- a variety of different sized beakers, test-tubes, boiling tubes, measuring cylinders or syringes for measuring volumes.

[Total: 12]

Mark Scheme (details)

The independent variable in this experiment is the glucose concentration while the dependent variable is average time taken for methylene blue to decolourise.

the likely outcome of the experiment/Trend

As glucose concentration increases, the average time taken for blue colour of methylene blue to disappear decreases. [1]

likely outcome of the experiment

- When glucose concentration increases, more hydrogen atoms are removed from glucose molecules by enzymes called dehydrogenases and passed to hydrogen acceptor NAD⁺ and FAD (co-enzyme) [1]

Method of measuring rate of respiration/rational of experiment

- Methylene blue mimic the action of NAD⁺ and FAD
Methylene blue turns from blue to colourless when it is reduced by hydrogen. [1]
- Respiration rate is measured by average time taken for methylene blue to turn from blue to colourless [1]

- Procedure** Set up a thermostatically controlled water bath at 37 °C.
- Using **dilution** to obtain at least five known glucose solution concentrations; and placed them in separate labelled test-tubes.

Concentration stated 0.1% - 10% (any reasonable range). Final volume stated. [1]

es	Volume of 10% glucose / cm ³	Volume of distilled water / cm ³	Concentration of glucose solution / %

- Label 6 boiling tubes A – E and F (control). Using a 5 cm³ syringe, add 5 cm³ of each glucose solution concentration to A – E and 5 cm³ of distilled water to F.
- Label another 6 boiling tubes A1 – F1. Using a 5 cm³ syringe, add 5 cm³ of yeast suspension to A1 – F1.
Use the same concentration of yeast suspension from the same stock for each reading and repeat as it affects the rate of respiration. This ensures initial concentration of the respiratory enzymes is kept constant. **(Controlled variable)**
- Using another 5 cm³ syringe, add 3cm³ of pH buffer to A1 – F1. Changes in pH may affect the active site configuration and therefore enzyme activity. Addition of pH buffer will keep pH constant. **(Controlled variable)**

6. Leave boiling tubes A – F and A1 – F1 in the thermostatically controlled water bath for 5 min **(equilibration/acclimatization time)** using stopwatch to time. [1]

Temperature must be kept constant as it affects enzyme activity, hence the rate of respiration. **(Controlled variable)**

7. Pour contents from A into A1, B into B1, C into C1, D into D1, E into E1 and F into F1.

8. **Add 3 drops of methylene blue into each tube and shake/stir each tube to mix the contents and place them back into the water bath. [1] (key step)**

The same amount and concentration of methylene blue used ensures consistency in the time taken for complete decolourisation. **(Controlled variable)**

Any 2 controlled variables – [1]

9. **Start the stopwatch and note the time taken for the blue colour to disappear from each tube. [1] (Method to measure)**

10. Repeat steps 1 to 9 to obtain a total of three readings, using freshly prepared yeast suspension, glucose solution and methylene blue each time. **(Reliability of results)**

11. Repeat the entire experiment (steps 1 to 10) twice using fresh yeast suspension, glucose solution and methylene blue of a different batch, to ensure reproducibility of results.
Replicate + Repeat [1]

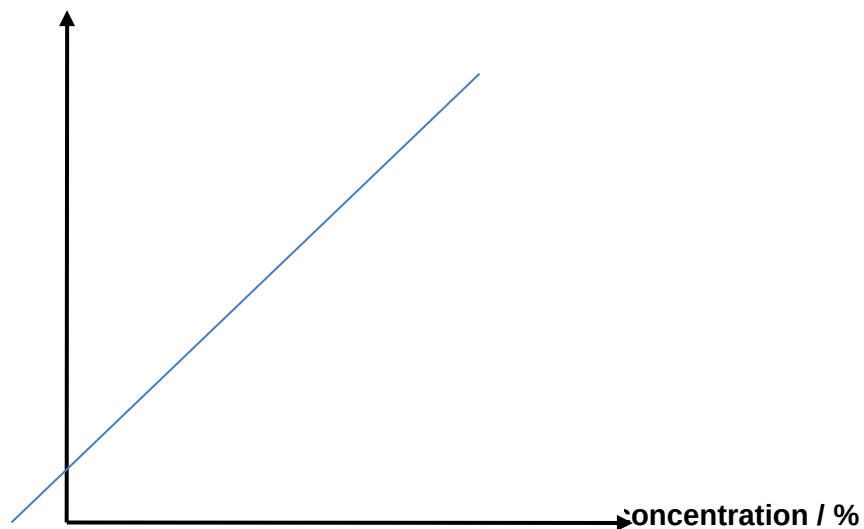
12. Plot a graph of rate of respiration / s^{-1} against glucose concentration / %. **[1] (with correctly plotted graph)**

Control

A control (test tube F) is set up which has 5 cm^3 of distilled water instead of glucose added to it and subjected to the same conditions as the rest of the tubes. To show that the colour change of methylene blue is due to the oxidation of glucose by yeast enzymes. [1]

Table showing rate of respiration with varying glucose concentration [1]

Glucose concentration / %	Time taken for blue colour to disappear / s				Rate of respiration / s^{-1}
	Reading 1	Reading 2	Reading 3	Average	

Graph of AVG rate of respiration/s⁻¹ against glucose concentration / %.**Rate of respiration / s⁻¹****Safety Precaution**

- Methylene blue can be a skin irritant – Wash under running water when in contact with skin.
 - Water bath may be hot and may scald. Wear gloves when handling beaker of hot water.
- [1] for 2 precautions

Fully Labelled Diagram [1]

Section E

Answer the question in this section on the answer paper provided.

Begin each part of the question on a new piece of answer paper.

5 Free-response question

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

(a) Describe the goals, benefits and ethical concerns of human genome project. [8]

(b) Explain the significance of genetic engineering in improving food quality. [6]

(c) Discuss the social and ethical implications of genetically modified crop plants. [6]

9 **(a)** Describe the goals, benefits and ethical concerns of Human Genome Project. [Total: 20] [8]

Goals of Human Genome Project: [Any 2 of the]

1. To determine the sequences of all the 3 billion DNA base pairs in the human genome and to store them in databases accessible to the public.
2. To identify and sequence all the 20,000 to 25,000 genes in the human DNA.
3. Using the gene sequences obtained, to construct a detailed genetic linkage map and physical map of the human genome to understand genetic basis of diseases.
4. The Project also aimed to sequence the genomes of several other organisms that are important to medical research, such as the mouse and the fruit fly.
5. Develop new tools to obtain and analyse the data and to make this information widely available. Computer programs or software would be developed and improved to analyze the data because the data are difficult to interpret without such programs. (OWTTE)
6. Develop genetic tests/screening for diseases and develop more efficient methods for DNA sequencing and sequence analysis and the transfer of these technologies to industry. (OWTTE)
7. Address the ethical, legal and social issues (ELSI) that may arise from the project.

[Max 3 of the above with elaboration]

Benefits of Human Genome Project (HGP) – Max 3 Marks

State + describe one example within category (include explanation to get full mark)

1. **Advancement in Genetic testing / Molecular Medicine**

[8]

- a) The HGP has allowed for discovery of genes/alleles/ associated with human diseases/ understanding genetics basis of disease; this allows for improved diagnosis of disease / genetic testing.
- b) Earlier detection of genetic predispositions to disease (eg. breast cancer, cystic fibrosis, Alzheimer's disease).
- c) It also allows scientists to design drugs to target a specific gene/protein.

2. Personalised Medicine / Pharmacogenomics

- a) The HGP allows scientists/doctors to know which genes/alleles affect a person's response to a drug since genetic differences affect the way we react to the same drug.
- b) It is now possible to tailor drugs/treatments to fit patient's genome for greater efficacy / avoiding dangerous side effects;

3. Improvement in Gene therapy

- a) Since we are able to have a understanding which genes/alleles are associated with which diseases, It is possible to use gene therapy to treat certain diseases since gene sequences are now readily available via databases,

4. Risk assessment of individuals upon exposure to toxic agents

- a) The HGP has allowed for **discovery of genes/alleles associated** with resistance/susceptibility to radiation/carcinogens. Individuals' genome can be used as a means of risk assessment to evaluate health risks of individuals who may be exposed to radiation or carcinogens. This will help to reduce the likelihood of heritable mutations.

5. Anthropology, Evolution, and Human Migration

- a) study **evolution through germline** mutations in lineages
- b) study population migration through matrilineal line based on female genetic inheritance or patrilineal line through Y chromosome mutations

6. Energy and Environmental Applications

- a) Use microbial genomics research to create new energy sources (biofuels)
- b) Use microbial genomics research to develop environmental monitoring techniques to detect pollutants
- c) Use microbial genomics research for safe, efficient environmental remediation

7. Advancement in Agriculture, Livestock Breeding (GMO)

- a) Creation of Disease-, insect-, and drought-resistant crops. E.g. BT corn,
- b) Creation of healthier, more productive, disease-resistant farm animals.

#

Ethical Concerns – 3 Marks

1. Privacy and confidentiality of genetic information. [Elaboration with one point from below]

- a) The issue of who owns genetic information – whether the individual has

complete control over who has access to his genetic information, or is access controlled by the company/researcher who carries out the genome sequencing, or even controlled the government.

- b) Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others.
- c) The issue of whether insurers / employers / courts / schools / adoption agencies / military may request for and have access to personal genetic information to discriminate people based on their genomes.

2. Psychological impact and stigmatization due to an individual's genetic differences.

- a) It is unclear how personal genetic information affects an individual and society's perceptions of that individual
- b) Will the genomic information lead to discrimination and affect members of minority communities

3. Reproductive issues including adequate informed consent for complex and potentially controversial procedures, use of genetic information in reproductive decision making, and reproductive rights.

- a) There is an issue of whether healthcare personnel are properly counseling parents about the risks and limitations of genetic technology (eg. with regards to the reliability of the genetic test, or whether the detected condition can be treated, and to help patients anticipate and deal with options to deal with the disease, if present, and whether relatives should be informed of the condition so that they can decide whether to test for the condition as well).
- b) To-be parents may have to make difficult decisions of whether to terminate pregnancy due to presence of genetic disorder (especially one for which there is currently no cure or treatment for).

4. Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions.

- a) The issue of whether testing should be performed when no treatment is available/treatment is extremely expensive and the patient cannot afford it, as diagnosing such a condition could lead to more anxiety and frustration.
- b) The issue of whether parents have the right to have their children tested for adult-onset diseases, as there is potential for conflict between a parent's choice and a child's welfare (eg. a parent refuses to consent to a test that is clearly in their child's best interest, or a parent who decides to pursue a genetic "enhancement" that involves significant risks for a child, or that may limit a child's life prospects)
- c) There is also the related issue of who has the right to determine whether newborns or others who are incapable of valid consent (eg. mentally incapacitated) should undergo genetic screening.
- d) The genetic tests may only indicate a probability and not a certainty of a particular polymorphism/allele being associated with a disease or condition. (There is difficulty in interpreting a positive result because some people who carry a disease-associated mutation never develop the disease.) Hence the genetic tests may not be reliable.

- (b) Explain the significance of Genetic Engineering in improving food quality.

Define Genetic Engineering [1]

Define genetic engineering:

the application of recombinant DNA technology to introduce genetic material/ foreign genes in order to alter the hereditary traits/ genetic makeup of a cell, organism, or population;

Genetically Modified Organism: Refers to an organism that has acquired one or more genes by artificial means. The genes may or may not be from the same species. OR
Transgenic organism: to describe organisms that had been genetically engineered to express a foreign gene from another species.

Improving Food Quality + Explain briefly [max 2 Marks each]

1. Improved Quality and Yield in Plants e.g. Bt corn [2]

- Development of plants resistant to insects / pests. E.g. BT corn express Bt toxins from the Bt toxin gene (Bt toxin gene is isolated from *Bacillus thuringiensis* and genetically engineered into corn crops).
- Growers use Bt corn as an alternative to spraying insecticides for control of corn borer. Consequently, farmers use less pesticides because BT corn express their own insecticidal proteins.
- This gene crystal proteins (Cry proteins) which acts as insect stomach poisons that must be eaten to kill the insect. Once eaten, the insect's own digestive enzymes activate the toxic form of the protein.
- The Cry proteins bind to specific receptors on the intestinal lining and rupture the cells causing death of the organism within 2 or 3 days.
- Bt maize has revolutionized pest control and many farmers have benefited financially.
- As this toxin is lethal to the pest but harmless to other animals, this Bt corn allows farmers to control pest infestations in order to reduce crop losses.

Others:

Development of Frost Resistance Plants. E.g. Frost-resistant Strawberries, edible vaccines

- *Development of crops with frost resistance. E.g. frost-resistant strawberries can be made by inserting the gene for antifreeze proteins from winter fishes into strawberry crops using recombinant DNA Technology.*
- *Development of bananas that contains human vaccines against infectious diseases such as hepatitis B*

2. Improved quality e.g. golden rice [2]

- Modification of crops by allowing them to produce additional vitamins / minerals. E.g. Golden rice has been genetically engineered to produce beta-carotene a precursor of vitamin A. Production of golden rice will help to provide nutritionally enriched foods particularly to those in developing countries.
- Vitamin A deficiency is the leading cause of preventable blindness in children and increases the risk of disease and death from severe infections.
- Rice grain, which serves as a food staple for much of the world do not contain vitamin A naturally.

- It was discovered that **geranyl geranyl diphosphate (GGPP)** found in rice seed can be a precursor to carotenoid production. **Beta-carotene** and other carotenes are natural **precursors** (inactive form) of **vitamin A**.
- Thus it is possible to genetically engineer a new breed of rice variety, **golden rice** which can **express the enzymes** necessary for the **conversion of GGPP to beta-carotene**.
- To engineer **golden rice**, genes coding for **phytoene synthase** (obtained from plant) and **phytoene desaturase** (obtained from bacteria) must be introduced into the rice plant cells. These enzyme-coding genes catalyze the biosynthesis of beta-carotene from precursor GGPP in the endosperm (edible part of the grain)
- A bacterium, ***Agrobacterium tumefaciens***, containing a **Ti plasmid**, was used to introduce all the **enzyme-coding genes**. OR another way of introducing DNA into plant cells is through DNA coated particles that are literally shot through the cell wall using a modified gun. This is commonly referred to as the use of a 'gene gun'.

3. Improved yield e.g. GM salmon [2]

- Production of GM salmon that grow and reach market size twice as fast as non-transgenic salmon for greater production of fish meat.
- Recombinant DNA composed of an **antifreeze promoter** from an ocean pout and a **growth hormone gene** from a Pacific Chinook salmon is synthesized. Fusing of a strong gene promoter such as the ocean pout antifreeze promoter leads to enhancement in the expression of the gene construct.
- The recombinant DNA is then introduced into fertilized eggs of Atlantic salmon. Subsequent selection and breeding led to development of the genetically modified salmon.
- Due to the year-round production of growth hormone (due to the antifreeze promoter), this allows for continuous feeding and growth of the GM salmon.
- The GM salmon is able to grow quicker in size while feeding more efficiently (less feed is consumed to reach a larger size).

Others:

Improving the nutrient and quality of food. E.g.

- Pharming of animals to produce vaccines or drugs for therapy / medicine. E.g. genetically engineering goats to produce anti-thrombin in their milk.
- For increase quality of meat, pigs have been genetically engineered to produce higher content of omega-3 fatty acid.

- (c) Discuss the social and ethical implications of GM crop plants.

[6]

Social Implications [Max 3 Marks]

1. Cost Savings for famers and cheaper foods for consumers {social}

- a) Growth of pest-resistant plants means that lesser crops are lost due to insect damage / diseases spread by insect vectors, therefore resulting in higher yield.
- b) Lesser insecticide / pesticide is used, therefore the farmers can save more money.
- c) Growth of herbicide-resistant plants means that farmers can spray herbicides to kill weeds without harming the crops. This results in reduction in competition from weeds, resulting in higher crop yield.
- d) Savings for famers translate into cheaper foods for consumers.

2. Malnutrition and reduce mortality rate {social}

- a) Enhanced quality of crop (e.g. Golden rice, expression of beta-carotene in endosperm in rice, which is converted into Vitamin A in the body when eaten) which helps to keep people healthy and to prevent malnutrition.
- b) Doing so will have a positive impact on public health, improving economic productivity, and individual well-being.

3. Rise of super-weeds lead to increase in expenses for Farmers {social}

- a) There is a risk of transgene transfer to closely-related non-crop species. For e.g. for herbicide-resistant GM crops, the use of herbicides can lead to the rise of super-weeds that are resistant to herbicides as weeds are grown alongside herbicide-resistant GM crops, due to crossing with the closely related GM plant;
- b) Herbicide acts as a selection pressure when a grower continues to use only one particular herbicide without any other herbicide modes of action, or doesn't use any other cultural practices. The resistant weed type continues to survive, mature and produce seed. Subsequent populations of the resistant biotype will continue to increase, reducing crop yields.
- c) Farmers would then have to resort to heavy / excessive usage of herbicide, which persist in the environment (i.e. runoff of chemicals into waterways), affecting the ecosystem.

4. Big biotechnology companies will monopolise the seed market, poor farmers not able to afford such seeds (Widening the gap between the rich and poor)

- a) GM suppliers could make farmers buy new seed every year. E.g., when farmers purchase a patented seed variety from Monsanto, they sign an agreement that they will not save and replant seeds produced from the seed they buy from Monsanto.
- b) Concerns that a few big biotechnology companies will dominate the world seed market.
- c) There are also concerns that GM crops might prove too expensive for poor farmers in developing countries to cultivate, thus widening the gap between the rich and poor.

- d) Using gene from animals in plant food foods may pose ethical or religious problems.
- e) Increasing dependence of developing nations on industrialized nations as the GM crops / seeds may be too expensive for the poor famers in developing countries;

5. Increase in antibiotic-resistance genes found in crops and gut bacteria

Concerns over effects in agriculture of crops interbreeding with closely related species making them resistant. E.g., there is a possibility that bacteria in our guts could pick up antibiotic-resistance genes found in any GM food. If this transfer occurs, it might exacerbate the already worrisome spread of disease-causing bacteria that are resistant to antibiotics.

Ethical Issues: Max 3 Marks

1. GM crops may have long-term effects on humans which are yet unknown. The FDA currently does not require safety assessments of GM foods to be done, in part due to the difficulty of conducting long term feeding trials.
2. There are concerns whether genetically modified foods would be acceptable to various religions or vegetarians. E.g. a vegetarian might not feel comfortable eating strawberries bearing antifreeze proteins from a winter fish / a person observing kosher dietary laws might be offended to know that a tomato he / she has eaten carried a gene isolated from pigs.
3. There are calls for GM food products to be labelled so that consumers can make informed choices. Currently, in the United States, food labeling of GM food is optional / voluntary, so consumers do not know they are consuming GM foods.
4. Critics argue that raising GM crops is an uncontrolled experiment with unknown consequences to the surrounding ecosystems, such as causing “genetic pollution” from the out-crossing of transgenic / GM organisms with wild populations. This mixing of genes and the formation of hybrids result in changes in the gene pools, causing a loss in biodiversity. This might lead to reduction in biodiversity and changes. Cite study showing reduced survival of Monarch caterpillars fed on milkweed dusted with Bt maize pollen.
5. Some critics oppose to GMOs on religious grounds, arguing that the act of altering genetic material goes against Nature and raises the idea of “playing of” god.

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